

Euler's Conjecture On Sums of Like Powers

In 1769 Leonhard Euler considered the following to be true: For all integers $n, k > 1$, if the sum of n k -th powers of positive integers is itself a k -th power then $n \geq k$

$$a_1^k + a_2^k + \cdots + a_n^k = b^k \quad \rightarrow n \geq k$$

COUNTEREXAMPLE TO EULER'S CONJECTURE ON SUMS OF LIKE POWERS

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A direct search on the CDC 6600 yielded

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5$$

as the smallest instance in which four fifth powers sum to a fifth power. This is a counterexample to a conjecture by Euler [1] that at least n n th powers are required to sum to an n th power, $n > 2$.

REFERENCE

1. L. E. Dickson, *History of the theory of numbers*, Vol. 2, Chelsea, New York, 1952, p. 648.

Lander, L. J., & Parkin, T. R. (1966) Bulletin of the American Mathematical Society, 72(6), 1079.

This was disproved using a computer in 1966:

Problem 1

1. Write a Python (no Numpy) program to compute the solution. Consider the following intervals for your computation:

$$20 \leq a_1 \leq 40$$

$$80 \leq a_2 \leq 100$$

$$100 \leq a_3 \leq 120$$

$$120 \leq a_4 \leq 140$$

$$140 \leq b \leq 160$$

2. Compute the execution time (using *timeit*

<https://docs.python.org/3/library/timeit.html>

Problem 2. A palindromic number reads the same both ways. For example, 9009 is a palindromic number. In fact, $9009 = 91 \times 99$. This is the largest palindromic number, a product of two 2-digit numbers. Find the largest palindrome made from the product of 2-digit and 3-digit numbers.

Problem 3. Consider the calendar. September, April, June, and November have 30 days. All other months except February have 31 days. February has 28 days in most years and 29 in leap years. Recall that a leap year occurs on any year evenly divisible by 4 but not on a century unless it is divisible by 400. The first of January in 1900 was a Monday. How many

Mondays fell on the second day of the month during the 20-th century (from 1/1/1901 to 12/31/2000)?

Problem 4. An 18-th century mathematician, Christian Goldbach, proposed the following conjecture: every odd composite number can be written as the sum of a prime and twice a square.

$$9 = 7 + 2 \times 1^2$$

$$15 = 7 + 2 \times 2^2$$

$$21 = 3 + 2 \times 3^2$$

$$25 = 7 + 2 \times 3^2$$

$$27 = 19 + 2 \times 2^2$$

$$33 = 31 + 2 \times 1^2$$

Find is the smallest odd composite that cannot be written as the sum of a prime and twice a square.

Problem 5. A Pythagorean triplet is a set of three natural numbers, $a < b < c$, for which $a^2 + b^2 = c^2$. For example, $3^2 + 4^2 = 5^2$ and therefore $(3, 4, 5)$ is a Pythagorean triplet. Exactly one Pythagorean triplet exists for which $a + b + c = 1000$. Find the product abc for such a triplet.